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Film laminating apparatus

Abstract

A film laminating apparatus is provided. The film laminating apparatus includes a roller and a platform arranged oppositely, and a blowing member. A first gap is defined between the roller and the platform. The flexible panel and the laminating film are interposed between the roller and the platform via the first gap. The flexible panel includes a display portion and a bonding portion, and a step difference is existed between the display portion and bonding portion such that a second gap is formed between the bonding portion and the platform. An air outlet of the blowing member is in communication with the second gap. The blowing member is configured to generate a positive pressure at the second gap to support the bonding portion of the flexible panel.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Phase of PCT Patent Application No. PCT/CN2021/140605 having International filing date of Dec. 22, 2021, which claims the benefit of priority of Chinese Application No. 202111501355.5 filed on Dec. 9, 2021. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD OF INVENTION

(2) The present disclosure relates to the technical field of manufacturing a apparatus, in particular to a film laminating apparatus.

BACKGROUND OF INVENTION

(3) As display technologies are continuously developing, flexible display panels have become more and more widely used. Among them, in a manufacturing process of the flexible OLED (organic light-emitting diode) panel, a rigid glass substrate is usually used as a support to manufacture each film layer of the OLED panel. Then the glass substrate is peeled off. As shown in both FIG. 1 and FIG. 2, when the OLED panel **1000** flows to the module section, a size of part of film layers **1002** (such as thin film encapsulation layer, polarizer, etc.) on the front side of the OLED panel **1000** is smaller than a flexible substrate **1001** of the OLED panel. Therefore, a step difference is formed between a display portion **1000A** and a bonding portion **1000B** of the OLED panel **1000**. When the glass substrate is peeled from the flexible substrate **1001**, the back plate film **2000** (BP) have to be laminated to the back of the flexible substrate **1001** as a film material stiffness support. Due to the existence of the step difference, the bonding portion **1000B** of the OLED panel **1000** is unsupported, which causes the formation of laminating bubbles **3000** at the position corresponding to the step difference after the back plate film **2000** is laminated, which affects product quality and process yield.

(4) Therefore, the existing technology has defeats, which need to be solved urgently.

SUMMARY OF INVENTION

Technical Problems

(5) The present disclosure provides a film laminating apparatus, which can solve the problem of laminating bubbles caused by a step difference between the display portion and the bonding portion of the flexible panel in the back plate film lamination process in the prior art.

Technical Solutions

(6) In order to solve the above-mentioned problems, the technical solution provided by the present disclosure is as follows:

(7) An embodiment of the present disclosure provides a film laminating apparatus for laminating a flexible panel and a laminating film. The film laminating apparatus comprises a roller and a platform arranged oppositely, and a blowing member; wherein the platform is configured to support the flexible panel and the laminating film, a first gap is defined between the roller and the platform, the flexible panel and the laminating film are interposed between the roller and the platform via the first gap, the laminating film is laminated on a surface of the flexible panel by rolling the roller; wherein the laminating film is disposed on a side of the flexible panel away from the platform, the flexible panel comprises a display portion and a bonding portion which is disposed on a side of the display portion, and a second gap is defined between the bonding portion of the flexible panel and the platform; and wherein an air outlet of the blowing member is in communication with the second gap, and the blowing member is configured to generate a positive pressure at the second gap to support the bonding portion of the flexible panel.

(8) Alternatively, in some embodiments of the present disclosure, the platform is provided with a plurality of blowing holes at a position corresponding to the bonding portion, and the blowing

holes are in communication with the air outlet of the blowing member.

(9) Alternatively, in some embodiments of the present disclosure, the platform comprises a first platform portion and a second platform portion which is positioned on a side of the first platform portion, and the display portion of the flexible panel is disposed with respect to the first platform portion, the bonding portion of the flexible panel is disposed with respect to the second platform portion; wherein a plurality of the blowing holes are defined by the second platform portion and penetrate through the second platform portion.

(10) Alternatively, in some embodiments of the present disclosure, the roller is connected to a first driving member, and the first driving member is configured to drive the roller to roll along a direction from the first platform portion to the second platform portion.

(11) Alternatively, in some embodiments of the present disclosure, when the roller rolls to the bonding portion, the blowing member is configured to generate the positive pressure at the second gap, and a position of the air outlet of the blowing member is synchronized with a laminating position of the roller on the laminating film.

(12) Alternatively, in some embodiments of the present disclosure, the film laminating apparatus further comprises a detecting member, the detecting member is positioned at an edge of the first platform portion close to the second platform portion, or the detecting member is positioned at a junction of the first platform portion and the second platform portion, and the detecting member is configured to detect whether the roller reaches a designated position.

(13) Alternatively, in some embodiments of the present disclosure, the detecting member is an infrared sensor and a direction of the emitted infrared light emitted by the infrared sensor is perpendicular to a surface of the first platform portion for contacting with the flexible panel.

(14) Alternatively, in some embodiments of the present disclosure, the film laminating apparatus further comprises a controller which controls initiation and termination of the first driving member and the blowing member; when the detecting member detects that the roller reaches the designated position, the blowing member is initiated under control of the controller.

(15) Alternatively, in some embodiments of the present disclosure, the blowing holes are arranged at intervals along a rolling direction of the roller, and the blowing member comprises a plurality of air knives, and the air knives are positioned with respect to the blowing holes.

(16) Alternatively, in some embodiments of the present disclosure, when the detecting member detects that the roller reaches the designated position, the controller controls sequential initiation of the air knives along the rolling direction of the roller, and an initiation speed of the air knives is synchronized with a rolling speed of the roller.

(17) Alternatively, in some embodiments of the present disclosure, the second platform portion comprises a platform body, a hollow area, and a plurality of baffles, the baffles are positioned in the hollow area perpendicular to a rolling direction of the roller, and the baffles are rotatably connected to the platform body; and wherein a third gap which is defined between adjacent two of the baffles is the blowing hole.

(18) Alternatively, in some embodiments of the present disclosure, each of the baffles is rotatably connected to the platform body via a shaft, and the baffles rotate with the rotation of the shafts.

(19) Alternatively, in some embodiments of the present disclosure, the film laminating apparatus further comprises a second driving member. Each of the shafts is connected with the second driving member. The second driving member is used for controlling the rotation of each of the shafts. The controller may further be used for controlling the initiation and termination of the second drive member.

(20) Alternatively, in some embodiments of the present disclosure, a shape of the blowing hole may be a slit shape, and an extending direction of the slit is perpendicular to a moving direction of the roller.

(21) Alternatively, in some embodiments of the present disclosure, a surface of the second platform portion facing the flexible panel is a first surface, the second platform portion comprises a platform

body and a plurality of hollow areas, and each of the hollow areas is a blowing hole. wherein the blowing holes penetrate through the second platform portion in a direction perpendicular to the first surface; or the blowing holes penetrate through the second platform along a direction at a predetermined inclination angle to relative the first surface.

(22) Alternatively, in some embodiments of the present disclosure, a plurality of suction holes are defined on the first platform portion and penetrate through the first platform portion, the film laminating apparatus further comprises a plurality of suction members connected to the suction holes, the suction members generate a negative pressure between the first platform portion and the display portion of the flexible panel via the suction holes, so as to suction the flexible panel on the first platform portion.

(23) Alternatively, in some embodiments of the present disclosure, the first platform portion and the second platform portion are an integral structure.

(24) Alternatively, in some embodiments of the present disclosure, the first platform portion and the second platform portion are arranged independently of each other. The second platform portion is movable relative to the first platform portion.

(25) Alternatively, in some embodiments of the present disclosure, the film laminating apparatus further comprises a third driving member. The second platform portion is connected to the third driving member. The third driving member is used for driving the second platform portion to move. The controller can further be used for controlling the initiation and termination of the third driving member.

(26) Beneficial Effects:

(27) Beneficial effects of the present disclosure are that: the film laminating apparatus provided by the present disclosure comprises a roller and a platform arranged oppositely, and a blowing member. The laminating film are interposed between the roller and the platform. A step difference is existed between the display portion and bonding portion such that a second gap is formed between the bonding portion and the platform. In the present disclosure, by providing the blowing member and an air outlet of the blowing member that is in communication with the second gap, the blowing member can generate a positive pressure at the second gap to support the bonding portion of the flexible panel, thereby solving a problem of formation of laminating bubbles caused by the step difference between the display portion and the bonding portion of the flexible panel and increasing product quality and process yield.

Description

DESCRIPTION OF DRAWINGS

(1) In order to more clearly describe the technical solutions in the embodiments of the present disclosure, the following will briefly introduce the accompanying drawings used in the description of the embodiments. Obviously, the accompanying drawings in the following description are only some embodiments of the present disclosure. For those skilled in the art, without creative work, other drawings can be obtained based on these drawings.

(2) FIG. 1 is a schematic diagram of a lamination process of an OLED panel and a back plate film in the prior art.

(3) FIG. 2 is a top view diagram of the film layers after the OLED panel and the back plate film are laminated in the prior art.

(4) FIG. 3 is a schematic diagram of a lamination process of a flexible panel and a laminating film provided by an embodiment of the present disclosure.

(5) FIG. 4 is a structural schematic diagram of a second platform portion and a blowing member provided by an embodiment of the present disclosure.

(6) FIG. 5 is a schematic structural diagram of one type of the second platform portion provided by

an embodiment of the present disclosure.

(7) FIG. 6 is a schematic result diagram of the results of another type of the second platform portion provided by an embodiment of the present disclosure.

(8) FIG. 7 is a schematic result diagram of yet another type of the second platform portion provided by an embodiment of the present disclosure.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(9) The technical solutions in the embodiments of the present disclosure will be clearly and completely described below combining with the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure, rather than all the embodiments. Based on the embodiments of the present disclosure, all other embodiments obtained by those skilled in the art without creative work shall fall within the protection scope of the present disclosure. In addition, it should be understood that the specific embodiments described herein are only configured to illustrate and explain the present disclosure, and are not configured to limit the present disclosure. In the present disclosure, if there is no opposite explanation, the used orientation terms, such as “upper” and “lower” generally refer to the upper and lower directions of the apparatus in actual use or working state, and specifically refer to the drawing directions in the drawings, while “inner” and “outer” refer to the contour of the apparatus.

(10) The embodiment of the present disclosure provides a film laminating apparatus, which can solve the problem of formation of laminating bubbles caused by the step difference between the display portion and the bonding portion of the flexible panel during the lamination process of the laminating film in the prior art.

(11) As shown in FIGS. 3-7, an embodiment of the present disclosure provides a film laminating apparatus, which comprises a roller 1 and a platform 2 arranged oppositely, and a blowing member 3. A first gap is defined between the roller 1 and the platform 2. The platform 2 is configured to support the flexible panel 100 and the laminating film 200. In the lamination process of the flexible panel 100 and the laminating film 200, the flexible panel 100 and the laminating film 200 are interposed between the roller 1 and the platform 2 via the first gap. The laminating film 200 is laminated on a surface of the flexible panel 100 by rolling the roller 1.

(12) The laminating film 200 is positioned on a side of the flexible panel 100 which is away from the platform 2. The flexible panel 100 comprises a display portion 100A and a bonding portion 100B positioned on a side of the display portion 100A. A step difference exists between the display portion 100A and the bonding portion 100B, such that a second gap is defined between the bonding portion 100B of the flexible panel 100 and the platform 2. An air outlet of the blowing member 3 communicates with the second gap, and the blowing member 3 is configured to generate a positive pressure at the second gap to support the bonding portion 100B of the flexible panel 100.

(13) By the above-mentioned design, the present disclosure can solve the problem of formation of laminating bubbles caused by the step difference between the display portion and the bonding portion of the flexible panel, thereby increasing product quality and process yield.

(14) The film laminating apparatus of the present disclosure will be described in detail below in conjunction with specific embodiments.

(15) The film laminating apparatus provided in the present embodiment can be used for laminating a flexible panel and a laminating film. Exemplarily, the film laminating apparatus can be used for laminating a flexible panel and a back plate. As shown in FIG. 3, the film laminating apparatus comprises a roller 1 and a platform 2 arranged oppositely. The platform 2 is configured to support the flexible panel 100 to be laminated and the laminating film 200. The roller 1 is arranged in parallel on top of the platform 2. Furthermore, a first gap is defined between the roller 1 and the platform 2. In the lamination process, the flexible panel 100 and the laminating film 200 are interposed between the roller 1 and the platform 2 via the first gap. The laminating film 200 is laminated on the surface of the flexible panel 100 by rolling the roller 1.

(16) The flexible panel **100** comprises a flexible substrate **101** and a functional film layer **102** which is disposed on the flexible substrate **101**. The flexible substrate **101** is prepared with driving circuits and connection lines. The functional film layer **102** is a stack of multiple film layers, comprising but not limited to a light-emitting layer, a thin film encapsulation layer, a polarizer, etc. The flexible panel **100** comprises a display portion **100A** and a bonding portion **100B** which is positioned on a side of the display portion **100A**. The area where the functional film layer **102** corresponds to is the display portion **100A**, while the area where is not covered by the functional film layer **102** is the bonding portion **100B**. In a front side (display surface) of the flexible panel **100**, a step difference is defined between the display portion **100A** and the bonding portion **100B**. The bonding portion **100B** is used for bonding to a flexible circuit board or a printed circuit board.

(17) When the flexible panel **100** is placed on the platform **2**, the front side of the flexible panel **100** faces toward the platform **2**. A back side of the flexible panel **100** is used for bonding to the laminating film **200**. Due to the step difference between the display portion **100A** and the bonding portion **100B**, a second gap is formed between the bonding portion **100B** of the flexible panel **100** and the platform **2**.

(18) The film laminating apparatus further comprises a blowing member **3**. An air outlet of the blowing member **3** is in communication with the second gap. The blowing member **3** is configured to generate a positive pressure at the second gap to support the bonding portion **100B** of the flexible panel **100**.

(19) Specifically, the platform **2** is provided with a plurality of blowing holes **221** at a position corresponding to the bonding portion **100B**. The blowing holes **221** are in communication with the air outlet of the blowing member **3**. The blowing member **3** blows air to the bonding portion **100B** of the flexible panel **100** via the blowing holes **221**, so that a positive airflow **M** is formed at the second gap to support the bonding portion **100B** of the flexible panel **100**. When the roller **1** rolls to the bonding portion **100B** of the flexible panel **100**, the blowing member **3** is used for generating a positive pressure airflow **M** at the second gap. A position of the air outlet of the blowing member **3** is synchronized with a laminating position of the roller **1** on the laminating film **200** to formed positive airflow **M** is configured to support the bonding portion **100B**.

(20) In an embodiment, the blowing member **3** may be positioned below the second platform portion **22**.

(21) In an embodiment, the platform **2** comprises a first platform portion **21** and a second platform portion **22** positioned on a side of the first platform portion **21**. In the present embodiment, the first platform portion **21** and the second platform portion **22** are integrally designed. The display portion **100A** of the flexible panel **100** is disposed corresponds to the first platform portion **21**. The bonding portion **100B** of the flexible panel **100** is disposed corresponds to the second platform portion **22**. The second platform portion **22** is evenly provided with a plurality of blowing holes **221** penetrating through the second platform portion **22**.

(22) The shape of the blowing hole **221** may be a slit shape. An extending direction of the slit is perpendicular to a moving direction of the roller **1**. It may also be a trumpet shape or a duckbill shape arranged at intervals, which is not limited here.

(23) The first platform portion **21** is provided with a plurality of suction holes **211** penetrating through the first platform portion **21**. The film laminating apparatus further comprises an adsorption member **4** positioned below the first platform portion **21** and connected (communicating) with a plurality of the adsorption holes **211**. The adsorption member **4** generates a negative pressure between the first platform portion **21** and the display portion **100A** of the flexible panel **100** via the adsorption holes **211** to suck and fix the flexible panel **100** on the first platform portion **21**.

(24) The film laminating apparatus also comprises a first driving member (not shown) and a detecting member (not shown). The roller **1** is connected to the first driving member. The first driving member is used for driving the roller **1** to roll in a first direction (as shown by an arrow in

FIG. 3), the first direction is a direction in which the first platform portion **21** faces toward the second platform portion **22**. The first driving member may be an air cylinder or a motor. The detecting member may be positioned at an edge of the first platform portion **21** close to the second platform portion **22**, or the detecting member may be positioned at a junction of the first platform portion **21** and the second platform portion **22**. The detecting member is used for detecting whether the roller **1** reaches the designated position.

(25) In an embodiment, the detecting member may be an infrared sensor and a direction of the emitted infrared light is perpendicular to a surface of the first platform portion **21** for contacting with the flexible panel **100**. When the roller **1** rolls to the top of the infrared sensor, the infrared sensor can detect the roller **1**. Certainly, in other embodiments, the detecting member may also be other types of sensors which is no limited herein.

(26) The film laminating apparatus further comprises a controller that controls initiation and termination of the first driving member and the blowing member **3**. When the detecting member detects that the roller **1** reaches a designated position, the controller controls the blowing component **3** to initiate. Exemplarily, the designated position may be an edge of the display portion **100A** close to the bonding portion **100B**, or a junction of the display portion **100A** and the bonding portion **100B**. The controller can also control an air output of the blowing member **3**, so that the air output of the blowing member **3** can be adjusted according to the stiffness of film material of the flexible panel **100**.

(27) As shown in FIG. 3, during the lamination process, the flexible panel **100** and a first end of the laminating film **200** (the end of the display portion **100A** away from the bonding portion **100B**) is interposed between the roller **1** and the platforms **2**. The roller **1** rolls in the first direction and presses the laminating film **200** onto the flexible panel **100**. Because the first platform portion **21** and the roller **1** can provide a clamping force to the display portion **100A** and the laminating film **200**, no bubbles are formed when the laminating film **200** is laminated to the display portion **100A**. When the detecting member detects that the roller **1** is rolled to a specified position, the controller controls the blowing member **3** to initiate. Meanwhile, the blowing member **3** blows air to the bonding portion **100B** via the blowing holes **221**. Because a positive airflow **M** formed at the second gap and the roller **1** can provide a certain clamping force to the bonding portion **100B** and the laminating film **200**, the junction between the display portion **100A** and the bonding portion **100B**, and also the bonding portion **100B** will not generate bubbles after laminating.

(28) Furthermore, the blowing member **3** may comprise a plurality of subunits. The controller may regionally control the air output of the blowing member **3** according to actual manufacturing process conditions. For example, with the pressing process of the roller **1**, the controller may control the subunits of the blowing member **3** at the corresponding position to reduce the air output after being pressed by the roller **1**.

(29) In an embodiment, as shown in FIG. 4, a plurality of the blowing holes **221** are arranged at intervals along a rolling direction of the roller **1**. The blowing member **3** comprises a plurality of air knives **31**. The air knives **31** is provided corresponding to a plurality of the blowing holes **221**, and each air knife **31** is connected with the controller. When the detecting member detects that the roller **1** reaches a designated position, the controller controls the air knives **31** to be activated sequentially along the rolling direction of the roller **1**. The initiation speed of the air knives **31** is synchronized with the rolling speed of the roller **1**, so that the positive airflow blown by the air knife **31** can provide a supporting force for the bonding portion **100B** at the pressing position of the roller **1**.

(30) In addition, with the pressing process of the roller **1**, the controller may control the air knives **31** at the corresponding positions to be sequentially closed along the rolling direction of the roller **1** after being pressed by the roller **1**. Because the bonding portion **100B** does not need a positive pressure airflow to provide supporting force at the position after being pressed by the roller **1**, the risk of “bulging” or “protrusion” that may occur at the corresponding position of the bonding

portion **10B** after being pressed by the roller **1** is avoided.

(31) In an embodiment, as shown in FIG. 5 and FIG. 6, a surface of the second platform portion **22** facing toward the flexible panel **100** is the first surface **22a**. The second platform portion **22** comprises a platform body **220** and a plurality of hollow areas. Each of the hollow areas correspondingly forms a blowing hole **221**. The blowing holes **221** penetrate through the second platform portion **22** in a direction perpendicular to the first surface **22a**. Alternatively, the blowing holes **221** penetrate through the second platform portion **22** along a direction at a predetermined inclination angle relative to the first surface **22a**. The predetermined inclination angle is between 30° and 150°. Exemplarily, as shown in FIGS. 5-6, the predetermined inclination angles are 45° and 135°. In the present embodiment, by arranging the blowing holes **221** to penetrate through the second platform portion **22** at different angles, the direction of airflow blowing from the blowing member **3** via the blowing holes **221** is adjusted according to the stiffness of the film material of the flexible panel **100**.

(32) In an embodiment, as shown in FIG. 7, the second platform portion **22** comprises a platform body **220**, a hollow area, and a plurality of baffles **222**. The baffles **222** are arranged in the hollow area perpendicular to the rolling direction the roller **1**. The baffles are rotatably connected to the platform body. A third gap which is defined between adjacent two of the baffles **222** is the blowing hole **221**. Specifically, each of the baffles **222** is rotatably connected to the platform body **220** via a shaft **223**. The baffles **222** can rotate with the rotation of the shafts **223**. The film laminating apparatus further comprises a second driving member. Each of the shafts **223** is connected with the second driving member. The controller can control the initiation and termination of the second drive member, and can individually control the rotation of each of the shafts **223** via the second drive member. Exemplarily, the controller controls the rotation of the shafts **223** so that the airflow blown from the blowing member **3** via the blowing holes **221** forms a sweeping airflow.

Specifically, the sweeping airflow moves from the junction of the display portion **100A** and the binding portion **100B** to the edge of the binding portion **100B** away from the side of the display portion **100A**. A moving speed of the sweeping airflow is synchronized with the rolling speed of the roller **1**, so that the sweeping airflow (positive pressure airflow) can provide a supporting force for the bonding portion **100B** at the position pressed by the roller **1**. The baffles **222** of the present embodiment form a structure similar to a louver. The rotation of the shafts **223** can be adjusted according to the stiffness of the film material of the flexible panel to flexibly adjust the direction of the air outlet via the blowing holes **221**.

(33) In addition, in the present embodiment, the controller can control the shafts **223** to rotate in turn at the corresponding positions after being pressed by the roller **1**, so as to drive a surface of the baffle **222** to be parallel to the surface of the second platform portion **22** (**22a** in FIG. 5), thereby closing the blowing holes **221** at the corresponding position after being pressed by the roller **1**.

(34) In an embodiment, the first platform portion **21** and the second platform portion **22** are arranged independently of each other. The film laminating apparatus further comprises a third driving member. The second platform portion **22** is connected to the third driving member. The controller can control the initiation and termination of the third driving member. The third driving member is used for driving the second platform portion **22** to move, so that the second platform portion **22** is movable relative to the first platform portion **22**. Using the present design, the blowing member **3** does not need to be provided with a plurality of air knives **31** along the moving direction of the roller **1**. Only one or more air knives **31** are needed to be positioned along the axial direction of the roller **1**. Exemplarily, the second platform portion **22** may move from the junction between the display portion **100A** and the bonding portion **100B** to the edge of the bonding portion **100B** away from the side of the display portion **100A**. The moving speed of the second platform portion **22** is synchronized with the moving speed of the roller **1**, so that the positive airflow blown by the air knives **31** can provide supporting force for the bonding portion **100B** at the pressing position of the roller **1**.

(35) In summary, in the present disclosure, by providing with a plurality of blowing holes at the position of the platform corresponding to the bonding portion of the flexible panel, and connecting the blowing holes to the blowing member, the blowing member generate a positive pressure at the second gap via the blowing hole to support the bonding portion of the flexible panel, thereby solving a problem of formation of laminating bubbles caused by the step difference between the display portion and the bonding portion of the flexible panel and increasing product quality and process yield.

(36) The embodiments of the present invention are described in detail above. Specific examples are used herein to illustrate the principles and implementation of the present disclosure. Descriptions of the above embodiments are only used for understanding the method and main ideas of the present disclosure. Meanwhile, for people skilled in the art, based on the idea of the present disclosure, will make changes in the specific implementation and the scope of application. In summary, the content of the present specification shall not be construed as limitation of the present disclosure.

Claims

1. A film laminating apparatus for laminating a flexible panel and a laminating film, the film laminating apparatus comprising: a roller; a platform arranged oppositely with respect to the roller; a blowing member positioned below the platform; and a detecting member; wherein the platform is configured to support the flexible panel and the laminating film, the platform comprises a first platform portion and a second platform portion positioned on a side of the first platform portion, the roller and the platform are separated by a first distance to form a first gap, the flexible panel and the laminating film are interposed between the roller and the platform via the first gap, and the laminating film is laminated on a surface of the flexible panel by rolling the roller; wherein the laminating film is disposed on a side of the flexible panel away from the platform, the flexible panel comprises a display portion and a bonding portion which is disposed on a side of the display portion, the display portion comes into contact with the platform and the bonding portion does not come into contact with the platform, and the bonding portion of the flexible panel and the platform are separated by a second distance to form a second gap; and wherein an air outlet of the blowing member is in communication with the second gap, and the blowing member is configured to generate a positive pressure at the second gap to support the bonding portion of the flexible panel; and wherein the display portion of the flexible panel is disposed with respect to the first platform portion, the bonding portion of the flexible panel is disposed with respect to the second platform portion, the detecting member is positioned at an edge of the first platform portion close to the second platform portion or the detecting member is positioned at a junction of the first platform portion and the second platform portion, and the detecting member is configured to detect whether the roller reaches a designated position.

2. The film laminating apparatus according to claim 1, wherein the platform is provided with a plurality of blowing holes at a position corresponding to the bonding portion, and the blowing holes are in communication with the air outlet of the blowing member.

3. The film laminating apparatus according to claim 2, wherein a plurality of the blowing holes are defined by the second platform portion and penetrate through the second platform portion.

4. The film laminating apparatus according to claim 3, wherein the roller is connected to a first driving member, and the first driving member is configured to drive the roller to roll along a direction from the first platform portion to the second platform portion.

5. The film laminating apparatus according to claim 4, wherein when the roller rolls to the bonding portion, the blowing member is configured to generate the positive pressure at the second gap, and a position of the air outlet of the blowing member is synchronized with a laminating position of the roller on the laminating film.

6. The film laminating apparatus according to claim 3, wherein a surface of the second platform

portion facing the flexible panel is a first surface, the second platform portion comprises a platform body and a plurality of hollow areas, and each of the hollow areas is a corresponding one of the blowing holes; and wherein the blowing holes penetrate through the second platform portion in a direction perpendicular to the first surface, or the blowing holes penetrate through the second platform along a direction at a predetermined inclination angle relative to the first surface.

7. The film laminating apparatus according to claim 3, wherein a plurality of suction holes are defined on the first platform portion and penetrate through the first platform portion, and the film laminating apparatus further comprises: a plurality of suction members connected to the suction holes, wherein the suction members generate a negative pressure between the first platform portion and the display portion of the flexible panel via the suction holes, so as to suction the flexible panel on the first platform portion.

8. The film laminating apparatus according to claim 3, wherein the first platform portion and the second platform portion are an integral structure.

9. The film laminating apparatus according to claim 2, wherein a shape of the blowing holes may be a slit shape, and an extending direction of the slit is perpendicular to a moving direction of the roller.

10. The film laminating apparatus according to claim 1, wherein the detecting member is an infrared sensor and a direction of the emitted infrared light emitted by the infrared sensor is perpendicular to a surface of the first platform portion for contacting with the flexible panel.

11. The film laminating apparatus according to claim 10, wherein the first platform portion and the second platform portion are arranged independently of each other, and the second platform portion is movable relative to the first platform portion.

12. The film laminating apparatus according to claim 11, wherein the film laminating apparatus further comprises: a third driving member, wherein the second platform portion is connected to the third driving member, and the third driving member is used for driving the second platform portion to move, wherein the controller is further used for controlling the initiation and termination of the third driving member.

13. The film laminating apparatus according to claim 1, wherein the film laminating apparatus further comprises: a controller which controls initiation and termination of the first driving member and the blowing member, wherein when the detecting member detects that the roller reaches the designated position, the blowing member is initiated under control of the controller.

14. The film laminating apparatus according to claim 13, wherein the blowing holes are arranged at intervals along a rolling direction of the roller, and the blowing member comprises a plurality of air knives, and the air knives are positioned with respect to the blowing holes.

15. The film laminating apparatus according to claim 14, wherein when the detecting member detects that the roller reaches the designated position, the controller controls sequential initiation of the air knives along the rolling direction of the roller, and an initiation speed of the air knives is synchronized with a rolling speed of the roller.

16. The film laminating apparatus according to claim 13, wherein the second platform portion comprises a platform body, a hollow area, and a plurality of baffles, the baffles are positioned in the hollow area perpendicular to a rolling direction of the roller, and the baffles are rotatably connected to the platform body, and wherein adjacent pairs of the baffles are separated by a third distance to form a third gap which corresponds to the blowing holes, respectively.

17. The film laminating apparatus according to claim 16, wherein each of the baffles is rotatably connected to the platform body via a shaft, and the baffles rotate with the rotation of the shafts.

18. The film laminating apparatus according to claim 17, wherein the film laminating apparatus further comprises: a second driving member, wherein each of the shafts is connected with the second driving member, and the second driving member is used for controlling the rotation of each of the shafts, wherein the controller is further used for controlling the initiation and termination of the second driving member.

19. The film laminating apparatus according to claim 1, wherein a step difference between the display portion and the bonding portion of the flexible panel defines the second distance that forms the second gap between the bonding portion of the flexible panel and the platform.
